A deterministic Hölder exponent for parabolic SPDEs with smooth transport noise

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Abstract

We consider scalar divergence-form parabolic Itô equations on the torus, with bounded measurable diffusion and progressively measurable transport coefficients that are smooth in space with deterministic derivative bounds. Under stochastic parabolicity, we prove that every variational solution with deterministic smooth initial data has a positive deterministic parabolic Hölder exponent on each compact positive-time strip. The Hölder norm may remain random and unbounded. The stochastic-flow transformation of Agresti, Sauerbrey and Veraar removes the noise, but its global ellipticity ratio can depend on the realization. We normalize the continuous inverse flow derivative by a constant affine map near each spacetime point. This gives uniform local energy-class constants on possibly random neighborhoods. Compactness preserves the common exponent, and composition with the inverse flow gives the assertion in the original coordinates.

1 Introduction and main result

For a scalar deterministic divergence-form parabolic equation, bounded measurable coefficients and a fixed ellipticity ratio give a common positive Hölder exponent. Smooth transport noise permits a stochastic change of coordinates that removes the noise, but the resulting global ellipticity ratio need not have a deterministic bound. Agresti, Sauerbrey and Veraar [1, Theorem 1.1 and Section 1.2] obtain pathwise Hölder regularity and ask whether its exponent can be chosen independently of the realization. Their Proposition 4.2 exhibits unbounded transformed ellipticity ratios; Proposition 4.3 gives uniform exponents under additional structure. The distortion of the transformed coefficients leaves open the regularity of the original solution.

Here we answer the deterministic-exponent question for the full smooth-noise class in AIM Problem 455. A large but finite coordinate distortion changes the Hölder constant, whereas a fixed affine change of coordinates preserves the exponent. At each point we use the inverse flow derivative itself for this affine change. Its continuity gives a neighborhood on which the normalized diffusion has deterministic bounds. The neighborhood can shrink with the sample, which is harmless when only the exponent must be deterministic. The stochastic flow and distributional Itô–Wentzell formula are taken from [1]. The deterministic regularity input is the parabolic De Giorgi class theorem of Liao [2]; we verify its energy inequalities for the nonsymmetric coefficients and bounded drift that occur here.

Write $\mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$, with its induced spatial distance dist. All function spaces below are real. Matrix norms are operator norms; other fixed matrix norms change only dimension constants. The

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space $H^{-1}(\mathbb{T}^d)$ is the dual of $H^1(\mathbb{T}^d)$ with respect to the L^2 pairing. On a filtered probability space, let $(W^k)_{k\geq 1}$ be independent standard Brownian motions. Consider the scalar Itô equation

$$\mathrm{d}u = \operatorname{div}(a\nabla u)\,\mathrm{d}t + \sum_{k\geq 1}(b_k\cdot\nabla u)\,\mathrm{d}W_t^k, \qquad u(0)=u_0. \tag{1}$$

The matrix a and the vector fields $b=(b_k)_{k\geq 1}$ are progressively measurable in (t,ω) and measurable in space. We assume deterministic constants $M<\infty$ and $\nu>0$ such that, almost everywhere,

$$|a| + \Big(\sum_k |b_k|^2\Big)^{1/2} \leq M, \tag{2}$$

$$\eta^\top\Big(a - \tfrac{1}{2}\sum_k b_k\otimes b_k\Big)\eta \geq \nu|\eta|^2 \qquad (\eta\in\mathbb{R}^d). \tag{3}$$

For every integer $j\geq 1$, b is spatially C^j as an $\ell^2(\mathbb{R}^d)$-valued map and

$$\sup_{t,\omega}\|b(t,\omega,\cdot)\|_{C^j(\mathbb{T}^d;\ell^2(\mathbb{R}^d))} \leq M_j < \infty \tag{4}$$

for fixed deterministic M_j. No symmetry of a is required.

We use the variational solution $u\in C([0,T];L^2)\cap L^2(0,T;H^1)$, almost surely. Thus, for every smooth spatial test φ, its L^2 pairing satisfies

$$(u(t),\varphi) = (u_0,\varphi) - \int_0^t (a\nabla u,\nabla\varphi)\,\mathrm{d}s + \sum_k\int_0^t (b_k\cdot\nabla u,\varphi)\,\mathrm{d}W_s^k.$$

The stochastic integrals are Itô integrals. A representative below agrees with this L^2-valued process at every time. For $0<\theta<1$, $C^{\theta/2,\theta}([\tau,T]\times\mathbb{T}^d)$ denotes the bounded continuous functions whose increments are bounded by a constant times $|t-s|^{\theta/2}+\operatorname{dist}(x,y)^\theta$.

Theorem 1.1. *Let $d\geq 2$, $T>0$, and let (2)–(4) hold. For every deterministic $u_0\in C^\infty(\mathbb{T}^d)$, the variational solution of (1) has, for every $0<\tau<T$, a representative satisfying*

$$\mathbb{P}(u\in C^{\gamma/2,\gamma}([\tau,T]\times\mathbb{T}^d)) = 1$$

for a deterministic $\gamma\in(0,1)$ depending only on d,ν,M. In particular it has the dependence allowed in AIM Problem 455. The Hölder norm need not be deterministically bounded. The vector fields may be noncommuting, time-random, and countably infinite in number.

The bounds M_j ensure the required regularity of the stochastic flow. They affect the random neighborhoods and constants in the proof, but not our choice of exponent. We do not assert moment bounds for the Hölder norm. We first prove the deterministic normalization lemma, then apply the stochastic change of coordinates and return to (1).

2 A local deterministic normalization

2.1 Bounded drift at unit scale

For $R>0$ write $K_R=(-R,R)^d$ and $Q_R=(-R^2,0)\times K_R$.

Lemma 2.1. *Suppose $w \in C_t L^2_{\mathrm{loc}} \cap L^2_t H^1_{\mathrm{loc}}$ is a scalar weak solution on Q_2 of*

$$w_t = \operatorname{div}(H\nabla w) + f \cdot \nabla w. \tag{5}$$

Assume $|H| \le \Lambda$, $\xi^\top H \xi \ge \lambda |\xi|^2$ and $|f| \le 1$ almost everywhere, where $\lambda > 0$ and $\Lambda < \infty$. There is $\alpha = \alpha(d, \lambda, \Lambda) \in (0,1)$ such that w has a $C^{\alpha/2,\alpha}$ representative on Q_1. If w has a continuous L^2 trace at time zero and its energy bounds hold to that time, this representative extends with the same exponent to the upper face of Q_1. Neither H nor f needs to be differentiable, and H need not be symmetric.

Proof. We check the energy-class hypotheses of [2, Section 1.1]. Fix a level $k \in \mathbb{R}$ and either truncation $z = (w-k)_+$ or $z = (w-k)_-$. For a nonnegative spatial cutoff ζ, use the signed test $z\zeta^2$ or $-z\zeta^2$, respectively. Coercivity and the full matrix norm bound give

$$\begin{aligned} \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\int z^2\zeta^2 + \lambda \int |\nabla z|^2 \zeta^2 &\le 2\Lambda \int |\nabla z|\zeta |\nabla \zeta| z + \int |f||\nabla z| z\zeta^2 \\ &\le \frac{\lambda}{2}\int |\nabla z|^2\zeta^2 + \frac{4\Lambda^2}{\lambda}\int z^2 |\nabla\zeta|^2 + \frac{1}{\lambda}\int z^2\zeta^2. \end{aligned}$$

For a time-dependent cutoff, add $\int z^2 |\zeta \zeta_t|$ on the right. All spatial integrals here are over the support of ζ. These tests are legitimate at the stated weak regularity: apply time Steklov averaging and smooth Lipschitz approximations to the signed truncations, integrate on a compact time interval, and pass to the limits. The averaged gradients and fluxes converge in L^2, their products in L^1, and the truncations converge in H^1 by the Sobolev chain rule. Time continuity gives the endpoint L^2 terms. No derivative falls on H or f.

For a translated cube $K_R(x)$, choose $\zeta = 1$ on $K_\rho(x)$, supported in $K_R(x)$, with $|\nabla\zeta| \le C_d/(R-\rho)$. All cubes lie in K_2, so $R - \rho \le 2$ and the last volume term is absorbed into a constant times $(R-\rho)^{-2}\int z^2$. With a time cutoff between $s-h$ and $s-\vartheta$, we obtain

$$\begin{aligned} &\operatorname*{ess\,sup}_{s-\vartheta<t<s} \int_{K_\rho(x)} z^2 + \int_{s-\vartheta}^{s}\int_{K_\rho(x)} |\nabla z|^2 \,\mathrm{d}t \\ &\qquad \le \Gamma\big((R-\rho)^{-2} + (h-\vartheta)^{-1}\big)\int_{s-h}^{s}\int_{K_R(x)} z^2 \,\mathrm{d}t, \end{aligned} \tag{6}$$

where $0 < \vartheta < h$ and $\Gamma = \Gamma(d, \lambda, \Lambda)$. Without a time cutoff, dropping the nonnegative gradient term gives

$$\operatorname*{ess\,sup}_{s-h<t<s} \int_{K_\rho(x)} z^2 \le \int_{K_R(x)} z(s-h)^2 + \frac{\Gamma}{(R-\rho)^2}\int_{s-h}^{s}\int_{K_R(x)} z^2 \,\mathrm{d}t. \tag{7}$$

The initial integral has coefficient exactly one. These inequalities hold for every admissible nested cylinder, every level, and both signs. Together with local time continuity, they put w in Liao's class $\mathfrak{B}_2(Q_2, \Gamma)$, as defined in [2, equations (1.1)–(1.2)].

By [2, Theorems 2.1 and 5.1], w is locally bounded and

$$\operatorname*{ess\,osc}_{Q_\rho(z_0)} w \le C \operatorname*{ess\,osc}_{Q_R(z_0)} w \,(\rho/R)^\alpha, \qquad 0 < \rho < R,$$

with C and $\alpha \in (0,1)$ depending only on $d, 2, \Gamma$. Here $Q_R(z_0)$ is the backward cylinder of radius R with top center z_0. Fixed small outer cylinders about points of Q_1 have a spatial and lower-time buffer in Q_2. The local boundedness estimate gives a common finite oscillation bound on these cylinders. For two nearby points, use a backward cylinder at the later time with radius larger than

their spatial separation and the square root of their time separation. Oscillation decay gives the parabolic α estimate; boundedness handles separated points.

For the upper face, first take cylinders with top time $-\varepsilon$. The outer buffers and the energy bound can be kept fixed as $\varepsilon \downarrow 0$, so the Hölder constants remain bounded. The representative extends continuously to time zero. Its L^2 limit equals the prescribed trace, by strong L^2 continuity. This proves the last assertion using only backward cylinders. □

2.2 A continuous matrix field

Lemma 2.2. *Let $K : [0,T] \times \mathbb{T}^d \to \mathrm{GL}(d,\mathbb{R})$ be continuous, and let S_0, c be bounded measurable coefficients such that*

$$|S_0| \le \Lambda_0, \qquad \xi^\top S_0 \xi \ge \nu |\xi|^2.$$

Suppose $v \in C([0,T];L^2) \cap L^2(0,T;H^1)$ solves

$$v_t = \operatorname{div}(A\nabla v) + c \cdot \nabla v, \qquad A = K S_0 K^\top. \tag{8}$$

Then, for every $0 < \tau < T$, v has a representative in $C^{\alpha/2,\alpha}([\tau,T] \times \mathbb{T}^d)$ for one $\alpha = \alpha(d,\nu,\Lambda_0) \in (0,1)$. Bounds on K, K^{-1} and c affect the Hölder constant, but not this exponent.

Proof. Fix $z_0 = (t_0, x_0)$ with $t_0 > 0$ and set $L = K(z_0)$. This is a constant invertible matrix. On a sufficiently small neighborhood U of z_0, continuity gives

$$P = L^{-1}K, \qquad |P - I| \le \tfrac{1}{2}.$$

Consequently

$$\xi^\top P S_0 P^\top \xi \ge \tfrac{\nu}{4}|\xi|^2, \qquad |P S_0 P^\top| \le \tfrac{9}{4}\Lambda_0 \le 4\Lambda_0. \tag{9}$$

Choose $r > 0$ so small that the affine cylinder

$$t = t_b + r^2 s, \qquad x = x_0 + rLy, \qquad (s,y) \in Q_2,$$

lies in U, lies in a torus coordinate chart, has lower time strictly positive, and satisfies $r|L^{-1}|\|c\|_\infty \le 1$. If $t_0 < T$, take $t_b > t_0$ with $0 < t_b - t_0 < \min\{r^2/2, (T-t_0)/2\}$; if $t_0 = T$, take $t_b = T$. Shrinking r makes all these requirements compatible.

For $W(s,y) = v(t_b + r^2 s, x_0 + rLy)$, the weak equation becomes

$$W_s = \operatorname{div}_y(H\nabla_y W) + f \cdot \nabla_y W, \qquad H = L^{-1}AL^{-\top} = P S_0 P^\top, \quad f = rL^{-1}c, \tag{10}$$

where coefficients on the right are evaluated at $(t_b + r^2 s, x_0 + rLy)$. The constant spatial Jacobian cancels in the weak formulation; the time factor r^2 cancels the two spatial derivative factors and leaves one factor r on the drift. By (9), Lemma 2.1 applies with $\lambda = \nu/4$ and $\Lambda = 4\Lambda_0$. These parameters do not involve L, r, c or the point z_0. Returning by the fixed affine map changes the Hölder constant and preserves the exponent. At $t_0 = T$ we use the upper-trace part of that lemma.

We have obtained the same exponent on a relative neighborhood of every point of $[\tau,T] \times \mathbb{T}^d$. Local continuous representatives agree on overlaps, since they agree almost everywhere with the energy solution. Time continuity identifies each time slice in L^2, including the upper trace, and spatial continuity then identifies it everywhere. A finite subcover has a positive Lebesgue number ℓ for the parabolic metric. Pairs at distance below ℓ are controlled in a common chart by the maximum of finitely many Hölder constants. For separated pairs, use $2\|v\|_\infty \ell^{-\alpha}$. Thus the global norm is finite. Only this norm, the charts and ℓ depend on the particular coefficients. □

3 Stochastic change of coordinates and proof of the theorem

3.1 The flow and the transformed equation

Put

$$B = \sum_k b_k \otimes b_k, \qquad S = a - \tfrac{1}{2}B, \qquad \Lambda_0 = M + \tfrac{1}{2}M^2. \tag{11}$$

Then $|S| \le \Lambda_0$ and $\eta^\top S \eta \ge \nu|\eta|^2$. Define the vector fields

$$\mu^i = \tfrac{1}{2}\sum_{k,j} b_k^j \partial_j b_k^i, \qquad q^i = \tfrac{1}{2}\sum_k (\operatorname{div} b_k) b_k^i. \tag{12}$$

The product rule and Cauchy–Schwarz in k bound their derivatives of every finite spatial order by finite combinations of M and the M_j. The sums are continuous bilinear contractions of spatially smooth ℓ^2-valued maps, and are progressively measurable.

Lift the periodic coefficients to $\mathbb{R}^d$ and solve the Itô flow equation

$$\mathrm{d}\Phi_t(x) = \mu(t, \Phi_t(x))\,\mathrm{d}t - \sum_k b_k(t, \Phi_t(x))\,\mathrm{d}W_t^k, \qquad \Phi_0(x) = x. \tag{13}$$

We use this Itô equation directly, so no time regularity or Stratonovich conversion of the random coefficient b is needed. The periodic lifts satisfy global spatial Lipschitz and growth bounds. Their higher spatial derivative bounds verify Assumption 2.5 of [1] with $k = 2$ and spatial Hölder parameter one: bounded third derivatives make second derivatives Lipschitz. For μ, derivatives of b through order four suffice. Theorems 2.3 and 2.6 of [1] give a C^2 diffeomorphism flow with jointly continuous spatial derivatives. Its inverse $\Psi_t = \Phi_t^{-1}$ is spatially Lipschitz and temporally $1/4$-Hölder on compact sets by [1, Proposition 2.8]. Remarks 2.4 and 2.7 there explicitly allow countably many Brownian motions and random coefficients merely measurable in time.

These assertions hold on one probability-one event. Periodicity and pathwise uniqueness give $\Phi_t(x + m) = \Phi_t(x) + m$ for integer m. One may first impose these identities for rational x and integer m at all times, then extend by continuity. The flow therefore descends to the torus. Define

$$F = D\Phi, \qquad K = F^{-1}, \qquad J = \det F. \tag{14}$$

The determinant is continuous and nonzero, and $J_0 = 1$, so $J > 0$. Compactness of $[0, T] \times \mathbb{T}^d$ gives finite pathwise bounds for F, K, J, J^{-1} and $\nabla \log J$. In particular K is a continuous invertible matrix field. All these bounds may depend on the realization.

Set $v(t, x) = u(t, \Phi_t(x))$ in the L^2 sense. The spatial chain rule and change of variables give

$$\begin{aligned} \|v(t)\|_2^2 &\le \|\det D\Psi_t\|_\infty \|u(t)\|_2^2, \\ \|\nabla v(t)\|_2^2 &\le \|D\Phi_t\|_\infty^2 \|\det D\Psi_t\|_\infty \|\nabla u(t)\|_2^2. \end{aligned}$$

Thus $v \in L_t^2 H^1$. It is also continuous in L^2. Indeed, split $u(t) \circ \Phi_t - u(s) \circ \Phi_s$ into the pullback of $u(t) - u(s)$ and the change of pullback of the fixed function $u(s)$. The first term tends to zero by the uniform pathwise pullback bound. For the second, approximate $u(s)$ in L^2 by smooth functions and use uniform continuity of Φ and the same bound. This includes both time endpoints.

We apply [1, Proposition 2.14], the distributional Itô–Wentzell formula. Its drift hypothesis is satisfied because $\operatorname{div}(a\nabla u) \in L_t^2 H_{\mathrm{loc}}^{-1} \subset L_t^1 H_{\mathrm{loc}}^{-1}$, and its noise hypothesis follows from

$$\sum_k |b_k \cdot \nabla u|^2 \le M^2 |\nabla u|^2.$$

The torus identity lifts to $\mathbb{R}^d$ by periodizing compactly supported test functions. The formula holds almost surely as an identity of continuous H^{-1}_{loc}-valued processes, at all times. With flow coefficients $\mu, -b$, its stochastic integrand is $(b_k \cdot \nabla u) \circ \Phi - ((b_k \cdot \nabla u) \circ \Phi) = 0$. Before spatial pullback, its remaining drift is

$$\operatorname{div}(a\nabla u) + \mu \cdot \nabla u - \sum_k b_k \cdot \nabla(b_k \cdot \nabla u) + \tfrac{1}{2} B : D^2 u = \operatorname{div}(S\nabla u) + q \cdot \nabla u. \tag{15}$$

For the signs and factors, put $C = 2\mu$. Distributionally,

$$\sum_k b_k \cdot \nabla(b_k \cdot \nabla u) = B : D^2 u + C \cdot \nabla u, \qquad \operatorname{div}(B\nabla u) = B : D^2 u + (C + 2q) \cdot \nabla u.$$

Substitution proves (15). Every differentiated coefficient in this calculation is a spatially smooth contraction of b; no derivative of a or pointwise Hessian of u is used.

For an L^2 vector flux h, the distributional change of variables is

$$(\operatorname{div} h) \circ \Phi = J^{-1} \operatorname{div}\big(JK(h \circ \Phi)\big). \tag{16}$$

To see the Jacobian factor, the distribution on the left pairs a test φ with $\operatorname{div} h$ against $(\varphi/J) \circ \Psi$. At $y = \Phi(x)$ its gradient is $K(x)^\top \nabla_x(\varphi/J)(x)$. Changing variables proves (16), first for smooth h and then for L^2 fluxes by density. Since $(\nabla u) \circ \Phi = K^\top \nabla v$, write

$$S_0 = S \circ \Phi, \quad q_0 = q \circ \Phi, \qquad A = KS_0K^\top, \qquad c = A^\top \nabla \log J + Kq_0. \tag{17}$$

Equations (15) and (16) yield

$$v_t = J^{-1} \operatorname{div}(JA\nabla v) + (Kq_0) \cdot \nabla v = \operatorname{div}(A\nabla v) + c \cdot \nabla v. \tag{18}$$

The transpose on A in (17) is needed when a is nonsymmetric. All factors in c are bounded pathwise. Smooth diffeomorphisms preserve spatial null sets; Fubini's theorem therefore preserves the bounds on S_0 almost everywhere in spacetime. This is an integrated H^{-1} identity as well as a distributional equation. We have taken no time derivative of J, K or b.

3.2 Return to the original coordinates

Proof of Theorem 1.1. On the common probability-one event above, the transformed equation (18) satisfies every hypothesis of Lemma 2.2. It follows that $v \in C^{\alpha/2,\alpha}([\tau, T] \times \mathbb{T}^d)$ with one deterministic

$$\alpha = \alpha(d, \nu/4, 4\Lambda_0) \in (0, 1), \qquad \Lambda_0 = M + \tfrac{1}{2}M^2,$$

where $\alpha(d, \lambda, \Lambda)$ is the choice in Lemma 2.1. The inverse-flow estimate gives, with a finite random constant,

$$\operatorname{dist}(\Psi_t(x), \Psi_s(y)) \le C_\omega\big(\operatorname{dist}(x, y) + |t - s|^{1/4}\big).$$

Since $u(t, x) = v(t, \Psi_t(x))$ in L^2 at every time, its continuous representative satisfies

$$\begin{aligned} |u(t, x) - u(s, y)| &\le C_\omega\big(|t - s|^{\alpha/2} + \operatorname{dist}(x, y)^\alpha + |t - s|^{\alpha/4}\big) \\ &\le C'_\omega\big(|t - s|^{\alpha/4} + \operatorname{dist}(x, y)^{\alpha/2}\big) \end{aligned}$$

on the compact strip. Thus $\gamma = \alpha/2$ proves the assertion. The terminal value is the actual L^2 trace, by the upper-trace argument in Lemma 2.1 and the continuity of the pullbacks.

The representative and the membership event can be chosen measurably. Apply a fixed sequence of periodic spatial mollifiers to the original L^2-valued process. On each regular path these mollifications converge uniformly on a compact strip to its unique continuous representative. Values at rational spacetime points and countable limits and suprema then determine its Hölder norm and the event that this norm is finite. No measurable choice of the random affine charts is needed. The same pathwise argument works on every positive-time strip; a countable exhaustion by rational lower times gives simultaneous interior regularity if desired. □

In the conclusion, $\gamma = \alpha/2$ is fixed by d, ν, M, and $C^{\gamma/2,\gamma}$ means temporal exponent $\alpha/4$ and spatial exponent $\alpha/2$. The assumptions are precisely (2)–(4) and the variational interpretation of (1). The local radii, covering constants and Hölder norm may depend on the realization; the exponent does not. The argument uses neither commutation of the transport fields nor volume preservation.

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